# Yunhan Qiao

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Ph.D. Candidate, Computer Science, Oregon State University

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## Research Interests

My research sits at the intersection of empirical software engineering, code comprehension, and human–AI collaboration. I study how students and experienced developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases, and whether faster, more correct work also means deeper understanding. I combine controlled experiments, screen-recorded behavioral analysis, prompt coding, EEG, and interviews to inform how computing education teaches and assesses GenAI-assisted development and how future AI coding agents and IDEs should be designed.

## Education

**Oregon State University** — Corvallis, OR, United States
*Ph.D. in Computer Science, School of EECS* — *Sep 2023 – Jun 2027 (expected)*
**Advisor**: Dr. Christopher Hundhausen at the SEAL Lab
**Research Areas**: Empirical Software Engineering, Code Comprehension, Computing Education, Human–AI Interaction

**Arizona State University** — Tempe, AZ, United States
*M.S. in Software Engineering* — *Aug 2021 – May 2023*
**Advisor**: Dr. Robert LiKamWa
**Thesis**: Networked System for Volumetric Athletic Coaching in Augmented Reality

**Guangxi University of Science and Technology** — Liuzhou, China
*B.E. in Software Engineering* — *Sep 2016 – Jun 2020*

**Southern Cross University** — Gold Coast, Australia
*B.E. in Information Technology* — *Sep 2017 – Jun 2020*

## Publications

*Journal Articles & Conference Papers*

**Yunhan Qiao**, Md Istiak Hossain Shihab, and Christopher Hundhausen. **A Systematic Literature Review of the Use of GenAI Assistants for Code Comprehension: Implications for Computing Education Research and Practice.** ***TOCE****: ACM Transactions on Computing Education*, 26(2), Article 24, 1–33, 2026. [PDF] [DOI]
**Selected by the TOCE editorial board for presentation at SIGCSE TS 2027.**

Ahsun Tariq, Phillip Conrad, Christopher Hundhausen, Sanjay Chandrasekaran, **Yunhan Qiao**, and Summit Haque. **Using SlackBot-Prompted Reflections to Promote Contribution Equality in Team Software Development Projects.** ***TOCE****: ACM Transactions on Computing Education*, 2026. [PDF] [DOI]

Md Istiak Hossain Shihab, Christopher Hundhausen, Ahsun Tariq, Summit Haque, **Yunhan Qiao**, and Brian Wise Mulanda. **The Effects of GitHub Copilot on Computing Students' Programming Effectiveness, Efficiency, and Processes in Brownfield Coding Tasks.** ***ICER'25****: Proceedings of the 2025 ACM Conference on International Computing Education Research*, Vol. 1, 407–420, Aug. 2025. [PDF] [DOI]

Manish Motwani, Aakash Kulkarni, **Yunhan Qiao**, Matthew Davis, and Ziyan Chen. **LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications.** ***AIxSE'25****: IEEE International Conference on AI x Software Engineering*, 59–66, 2025. [PDF] [DOI]

*Under Review*

**Yunhan Qiao**, Md Istiak Hossain Shihab, Summit Haque, and Christopher Hundhausen. **Faster but Not Wiser: GitHub Copilot Decouples Programming Performance from Code Comprehension in Brownfield Tasks.** ***TOCE****: ACM Transactions on Computing Education*, under review. [arXiv]

**Yunhan Qiao**, Summit Haque, and Christopher Hundhausen. **Beyond the Prompt: Linking What Developers Ask, Do, and Understand with Coding Agents.** ***SANER'27****: IEEE International Conference on Software Analysis, Evolution and Reengineering*, under review. [arXiv]

*Under Submission*

O. Schutz, **Yunhan Qiao**, Fatima A. Moussaoui, Rudrajit Choudhuri, and Anita Sarma. **Heart and Mind in the Loop: Why We Vibe When We Work with AI.** ***CHASE'27****: International Conference on Cooperative and Human Aspects of Software Engineering*, under submission.

## Teaching Experience

**Graduate Teaching Assistant, School of EECS** — Oregon State University
*Undergraduate software engineering courses* — *2024 – current*
**Courses**: CS 362 Software Engineering II; CS 461 Senior Software Engineering Project; CS 467 Online Capstone Project; CS 492 Mobile Software Development

## Professional Experience

**Software Engineering Intern, Mocha Software** Nanjing, China
*Software development* *Aug 2020 – Nov 2020*

Developed a conference system for China Mobile Migu Culture & Technology Group with Spring, Spring MVC, and MyBatis, using Ajax to cut page rendering time from 1 s to 0.5 s and dynamic SQL with Redis-backed SSO sessions to reduce search load.

## Awards & Scholarships

**Scholarly Presentation Award**: Oregon State University *2026*
**EECS Graduate Scholarship**: Oregon State University *2023 – current*
**Merit Award**: Guangxi University of Science and Technology *2018*

## Academic Service

SIGCSE TS 2027, Main Track, Program Committee

SANER 2027, Short Papers and Posters Track, Program Committee

## Skills & Coursework

**Research Methods**: Qualitative Analysis, Quantitative Analysis, Interviews, Survey Design
**Programming**: Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
**Frameworks & Tools**: React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, LaTeX, Figma; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
**Coursework**: Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis

## Languages

English (Fluent), Mandarin Chinese (Native)